Problems in Ordinary Differential Equations

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1 Introduction

This document contains solutions to some problems in ODE, from the textbooks of Perko; Hirsch, Smale and Devaney; and Hydon

2 Perko

Chapter 1

Problem Set 5, Problem 3 From the Theorem, we must have $|b| > 2$ and $a \in (-\frac{b^2}{2}, -2)$

Problem Set 9, Problem 5 We have that

$$E^s = \mathbb{R}^n \cap \bigoplus_{\mathrm{Re}\lambda<0} G(\lambda)$$

a Follows from Theorem 2

b Follows from Theorem 3

Chapter 2

Problem Set 1

1. There are 3 parts

   (a)
   $$Df(\mathbf{x}) = \begin{bmatrix} 1+x_2^2 & 2x_1x_2 \\ 2x_1 & -1+2x_2 \end{bmatrix}$$
   $$Dg(\mathbf{x}) = \begin{bmatrix} 1+x_2^2+x_3^2 & 2x_1x_2 & 2x_1x_3 \\ -1+x_2x_3 & 1-x_3+x_1x_3 & -x_2+x_1x_2 \\ -2x_1 & 1 & 1 \end{bmatrix}$$
   (b) $f^{-1}0 = (0,0),(0,1)$ and $g^{-1}0 = (0,0,0),(0,-1,1)$

(c) $D^2 f(\mathbf{x}) = \begin{bmatrix} 0 & 2x_2 \\ 2x_2 & 2x_1 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$

2. There are 2 parts
   (a) this function is C^1 on the punctured plane
   (b) set $E = \{(x, y) : x > -1, y > -2, x \neq 1\} \setminus \{(0, 0)\}$
3. Define $x(t) = \begin{cases} \frac{t^2}{4} \text{ for } t \geq 0 \\ 0 \text{ else} \end{cases}$
   Then, 4 different solutions are $x, (t \to -x(-t)), (t \to x(t) - x(-t))$ and 0
4. Define $x(t) = \frac{2}{\sqrt{1-8t}}$ for $t < \frac{1}{8}$
5. The function $x(t) = \sqrt{t}$ is a solution for $t > 0$.

Problem Set 2

1. We have $u_1(t) = 1 + t$, $u_2(t) = 1 + t + t^2 + \frac{t^3}{3}$, and $u_3(t) = 1 + t + t^2 + t^3 + \frac{2t^4}{3} + \frac{t^5}{3} + \frac{t^6}{9} + \frac{t^7}{63}$. It is easy to use induction to show that this pattern continues.
2. We easily get that $u_k(t) = (\sum_{i=0}^{k} \frac{(At)^k}{k!})x_0$

Problem Set 4

1. There are 5 parts
   (a) Assume $x_0 > 0$. Define $x : (-\infty, \frac{1}{x_0}) \to \mathbb{R}$ by $x(t) = \frac{x_0}{1-x_0 t}$. Since this function is unbounded near $\frac{1}{x_0}$, this is the maximal interval by Theorem 1.
   (b)
   - We have the function $x : (-1, 1) \to (-\frac{\pi}{2}, \frac{\pi}{2})$ given by $x(t) = \arcsin(t)$. This function satisfies the IVP.
   - Suppose there is a function $y : (-\alpha, \beta) \to E = \mathbb{R} \setminus \{(n + \frac{1}{2})\pi\}$ satisfying the IVP. Then by Theorem 1 $y(t) = x(t)$ for all $t \in (-1, 1)$.
     Suppose $\beta > 1$. Then $y(1) = \lim_{t \to 1^-} y(t) = \frac{\pi}{2}$, which is a contradiction.
     Similarly $\alpha \geq -1$

   (c) Define $x(t) - 2\tanh(2t)$
   (d) Define $x(t) = \frac{x_0}{\sqrt{1-2x_0^2 t}}$ for $t < \frac{1}{2x_0^2}$. Since this function is unbounded near $\frac{1}{2x_0^2}$, this is the maximal interval by Theorem 1.

(e) Define $x : (-\infty, \frac{1}{y_1}) \to E = (\mathbb{R} \setminus \{0\}) \times \mathbb{R}$ by $x(t) = (\frac{y_1}{1-y_1 t}, (y_2 - 1 + \frac{1}{y_1})e^t + t + 1 - \frac{1}{y_1})$. Since $x(t)$ is unbounded near $\frac{1}{y_1}$, by Theorem 1 this is the maximal interval.

2. Here are the parts

   (a) This is the same as the previous problem, with a particular initial value.

   (b)
   - Define $x : (-1, 1) \to E = (\mathbb{R} \setminus \{0\}) \times \mathbb{R}$ by $x(t) = (\sqrt{1+t}, \frac{1}{1-t})$
   - The interval cannot be extended beyond 1 because $x(t)$ is unbounded near 1
   - The interval cannot be extended beyond -1 because $\lim_{t \to -1^+} x(t) = (0, \frac{1}{2}) \notin E$

   (c) Define $x : (-1, \infty) \to E = (\mathbb{R} \setminus \{0\}) \times \mathbb{R}$ by $x(t) = (\sqrt{1+t}, \frac{2}{3}\sqrt{1+t}(1+t) + \frac{1}{3}$. This cannot be extended beyond -1 because $\lim_{t \to -1^+} x(t) = (0, \frac{1}{3}) \notin E$

3.

4. We have the map $f : E \to E$ given by $f(x_1, x_2, x_3) = (-\frac{x_2}{x_3^2}, \frac{x_1}{x_3^2}, 1)$ where $E = \mathbb{R} \times \mathbb{R} \times (\mathbb{R} \setminus \{0\})$.

   We have the covering map

   $$\text{cyl} = \text{pol} \times \text{id}_{\mathbb{R} \setminus \{0\}}$$

   where $\text{pol} : \mathbb{R} \times (0, \infty) \to \mathbb{R}^2 \setminus \{0\}$ is the smooth surjective covering map given by $\text{pol}(\theta, r) = (r\cos\theta, r\sin\theta)$

   Suppose we have a differentiable path $x : [0, 1] \to E$ such that $x' = f \circ x$ and $x(0) = \text{cyl}(p_0)$.

   By Lemma 54.1 of Munkres, we get $y : [0, 1] \to \mathbb{R} \times (0, \infty) \times (\mathbb{R} \setminus \{0\})$ with $y(0) = p_0$ and $x = \text{cyl} \circ y$

3 Hydon, Chapter 1

1. The solutions are $y = mx$. To see that these are the only solutions, suppose $y : (a, b) \to \mathbb{R}$ is a solution, where $0 \notin (a, b)$. Define $f(t) = \frac{y(t)}{t}$. Then $f'(t) = 0$.

   Some symmetries of this ODE are:

   - rotations by the matrices $R_\theta \in SO_2$
   - the reflection $\text{id} \times (-\text{id})$

2. We have $f : (\mathbb{R} \setminus \{0\}) \times \mathbb{R} \to \mathbb{R}$ given by $f(t,x) = \frac{1-x^2}{t}$

   We have the linear map $T_\epsilon = e^\epsilon \mathrm{id} \times \mathrm{id}$

   An arbitrary solution is a map $x : 0 \notin (a,b) \to \mathbb{R}$ such that $x'(t) = f(t,x(t))$

   The image of this curve under T_ϵ is the map $e^\epsilon(a,b) \to \mathbb{R}$ given by $t \to x(te^{-\epsilon})$

   It is clear that these sets are the same.

3. We have the function $f : U \to \mathbb{R}^2$ given by $f(x,y) = 1, \frac{y^3+x^2y-x-y}{xy^2+x^3+y-x}$ where $U \subseteq \mathbb{R}^2$ is the set where the denominator is nonzero.

4 Hirsch-Smale-Devaney, Chapter 8

1. (a) The equilibrium point $x_{m,n}$ is given by $x_{m,n} = \pi(n, m+\frac{1}{2})$.

   We have that $DF_{x_{m,n}} = \begin{pmatrix} (-1)^n & 0 \\ 0 & (-1)^{m+1} \end{pmatrix}$

   (b) The only equilibrium point is the origin.

   We have that $DF_0 = 0$

   (c) The only equilibrium point is the origin.

   We have that $DF_0 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

   (d) The equilibrium points are given by $x_s = (s,0)$

   We have that $DF_{x_s} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

   (e) The only equilibrium point is the origin.

   We have that $DF_0 = 0$

2. Consider the diffeomorphism $(x,y,z) \to (x + \frac{1}{3}y^2, y, z + y^2)$